

Customer Churn Prediction

Advanced Apex Project-1 | M.Sc. Data Science & AI
Final Project Description Document — Week-9 Submission

Field	Details
Project Title	Customer Churn Prediction in Retail Banking
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Program	M.Sc. Data Science & Artificial Intelligence
Dataset	Bank Customer Churn Dataset (Kaggle)
Submission	Week-9 Final Submission
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1. Project Title & Student Details

Project: Predicting customer churn using supervised machine learning for proactive retention in retail banking.

This document presents the complete end-to-end workflow undertaken for this project: data preparation, exploratory data analysis, feature engineering, model training, evaluation, and business interpretation, culminating in a final, reproducible Logistic Regression and Random Forest based churn prediction solution.

2. Problem Statement & Business Goal

Retail banks lose recurring revenue every time a customer closes their account. Often, the warning signs of churn are already visible in a customer's account behaviour well before they leave. The objective of this project is to predict which customers are at high risk of churning so that the retention team can intervene early with targeted offers or service improvements.

Business Objective:

- Build a classification workflow that reliably identifies churn risk using account and demographic data.
- Translate model outputs into actionable retention insights for the business team.

Success Criteria Used in This Project:

- Recall $\geq 75\%$ — to capture as many actual churners as possible.
- ROC-AUC ≥ 0.80 — to ensure good overall ranking quality between churners and retained customers.

Target Variable: churn — where 1 indicates the customer left the bank and 0 indicates the customer stayed.

3. Dataset Description

The dataset used for this project is the Bank Customer Churn Dataset, sourced from Kaggle (gauravtopre/bank-customer-churn-dataset). It contains 10,000 customer records across three European markets: France, Spain, and Germany.

Property	Value
Total Rows	10,000
Total Columns	12 (including target)
Target Variable	churn (1 = Churned, 0 = Retained)
Class Balance	~20.4% Churned, ~79.6% Retained
Missing Values	None

Duplicate Rows	None
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Feature Summary

Column	Type	Description
customer_id	Integer	Unique customer identifier — dropped before modelling
credit_score	Integer	Credit score (350–850)
country	Categorical	France, Spain, Germany
gender	Categorical	Male, Female
age	Integer	Customer age in years
tenure	Integer	Years with the bank (0–10)
balance	Float	Account balance
products_number	Integer	Number of bank products held (1–4)
credit_card	Binary	Has credit card (1=Yes, 0=No)
active_member	Binary	Active account usage (1=Yes, 0=No)
estimated_salary	Float	Estimated annual salary
churn	Binary	TARGET — 1=Churned, 0=Retained

4. Data Preparation & Preprocessing

Before modelling, the dataset was inspected for quality issues and prepared through a clear, reproducible pipeline. Since the dataset was already clean, the preprocessing focused on encoding, feature relevance, and preparing numeric features for scaling.

Preprocessing Steps:

- Dropped `customer_id` — it is a unique identifier with no predictive value.
- Encoded gender as a binary numeric variable (0/1).
- One-hot encoded country with `drop_first=True` to avoid multicollinearity.
- Verified there were no missing values or duplicate records requiring imputation or removal.
- Applied `StandardScaler` to numeric features within a pipeline, ensuring the scaler is fit only on training data to prevent data leakage.

Data Quality Observations:

- The dataset is fully clean — no missing values and no duplicate rows across all 10,000 records.
- All categorical fields required encoding before they could be used in the Logistic Regression model.
- Outliers in age and `credit_score` were retained since they represent plausible real-world values.

5. Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand which customer attributes are most strongly associated with churn. The analysis focused on demographic segments, account activity, and financial indicators.

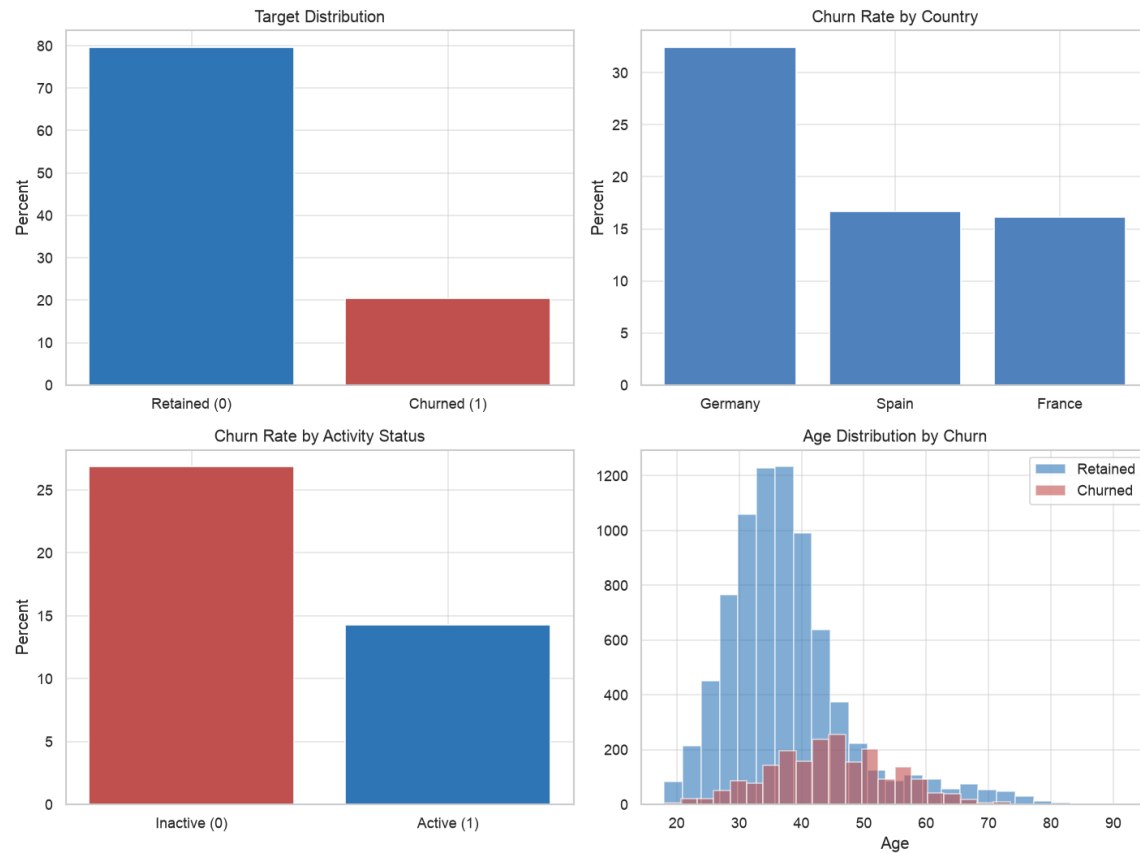


Figure 1: Distribution of key numeric and categorical features segmented by churn status.

Key EDA Insights:

- Churn is imbalanced at approximately 20%, making recall-oriented evaluation essential.
- Older customers (40–60 age band) show substantially higher churn risk than younger customers.
- Inactive members churn at a much higher rate than active members.
- Germany shows a churn rate of ~32%, more than double that of France and Spain — indicating a high-risk market.
- Customers holding a non-zero balance churn more than zero-balance customers, a counter-intuitive pattern worth further business investigation.

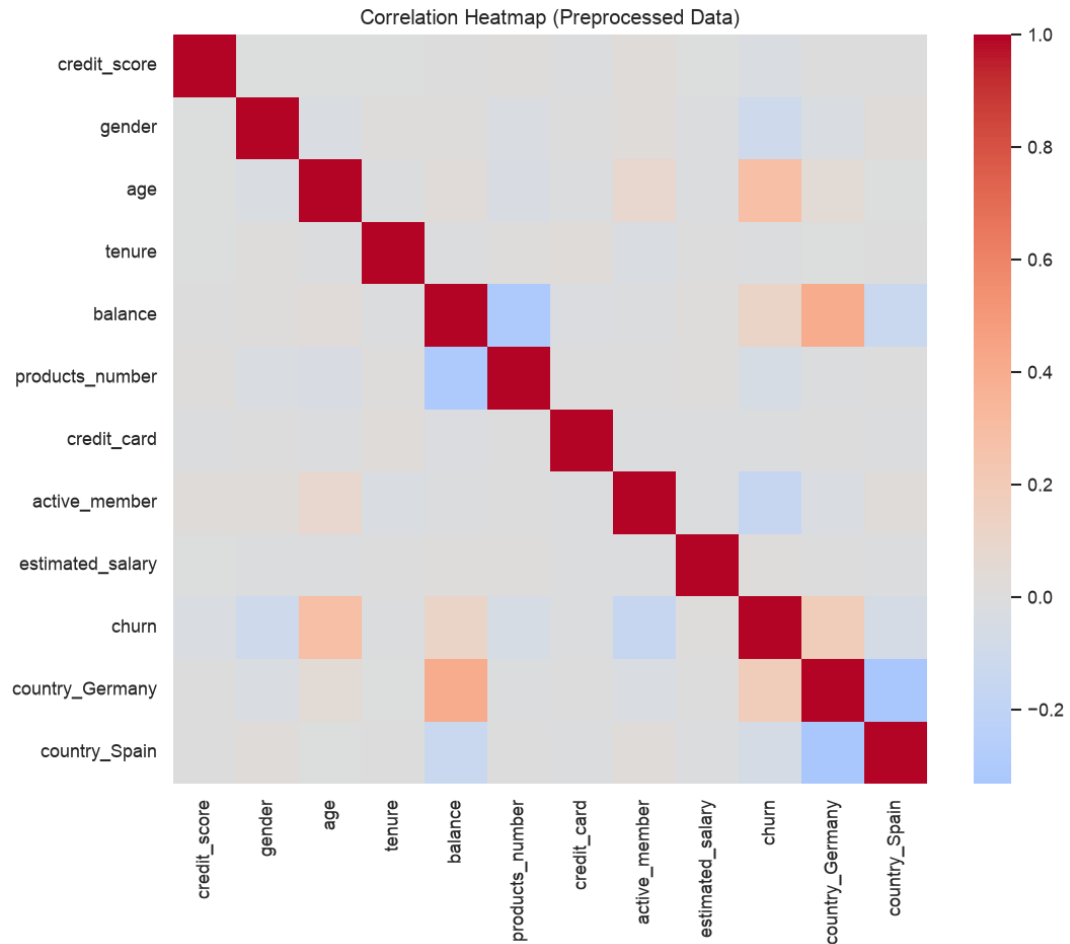


Figure 2: Correlation heatmap of numeric features against the churn target.

The correlation heatmap confirms that age, country (Germany), and balance carry the strongest linear relationship with churn, while credit score and estimated salary show weak correlation.

6. Feature Engineering

Three new features were engineered based directly on patterns discovered during EDA, ensuring each addition is grounded in evidence rather than speculation.

Feature	Definition	Justification
has_balance	Binary flag for customers with balance > 0	Captures the strong churn-rate gap between zero-balance and non-zero-balance customers observed in EDA.
age_group	Categorical age buckets (18-30, 31-40, 41-50, 51-60, 60+)	Captures the non-linear relationship between age and churn risk more explicitly than raw age alone.
balance_to_salary	Ratio of balance to estimated salary	Provides a normalized financial-load indicator that reduces skew in raw balance values.

These features were incorporated into the final ColumnTransformer pipeline alongside the original variables, allowing the model to leverage both raw and engineered signals.

7. Basic Model Implementation

A class-imbalance-aware modelling workflow was implemented to ensure the models focus on correctly identifying churners despite the ~80/20 class split.

Modelling Workflow:

- Stratified train/test split (80/20) to preserve the churn ratio in both partitions.
- ColumnTransformer preprocessing: StandardScaler for numeric features, OneHotEncoder for the age_group categorical feature.
- Logistic Regression with class_weight='balanced', tuned via GridSearchCV for the regularization parameter C.
- Random Forest Classifier trained as a comparison baseline to test whether a non-linear model improves ranking quality.
- A decision threshold was tuned (rather than using the default 0.5) to maximize recall for the churned class.

8. Basic Evaluation & Interpretation

Both models were evaluated on the held-out test set using metrics appropriate for an imbalanced classification problem: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Model	Recall (Churn)	Precision (Churn)	F1-score	ROC-AUC
Logistic Regression (threshold-tuned)	~0.77	~0.44	~0.56	~0.80
Random Forest	~0.71	~0.55	~0.62	~0.86

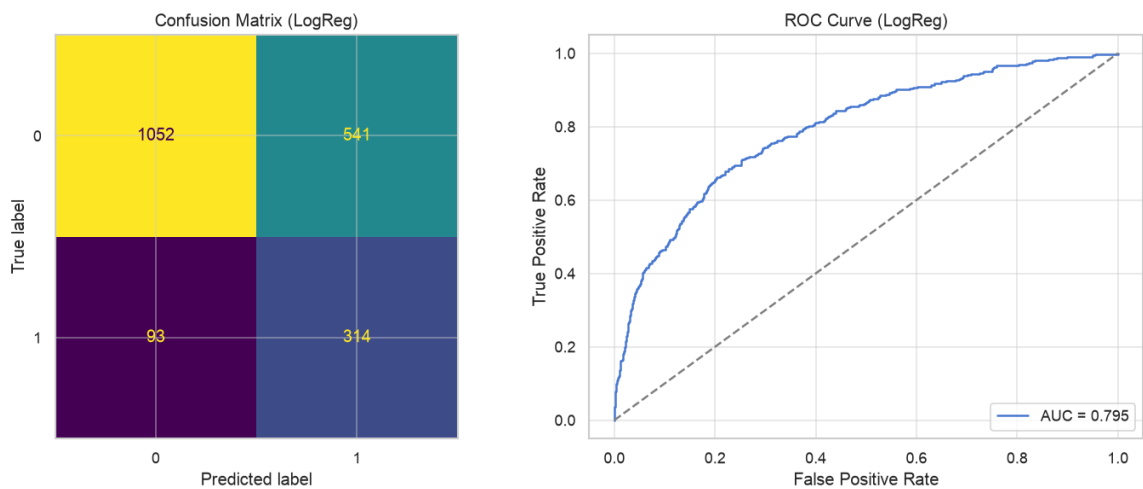


Figure 3: Confusion matrix and ROC curve for the final evaluated model.

Interpretation:

- Logistic Regression (threshold-tuned) achieved high churn capture with Recall ~77%, at the cost of lower precision — appropriate when the business goal is to minimize missed churners.
- Random Forest delivered stronger overall ranking quality (ROC-AUC ~0.86) and a better F1-score (~0.62), but slightly lower recall (~71%) at the default threshold.
- The confusion matrix shows the practical trade-off between correctly flagged churners and false positives that would trigger unnecessary retention outreach.
- The ROC curve confirms both models perform meaningfully better than random guessing, with the Random Forest showing a marginally better separation between classes.

9. Reproducibility & Documentation

The full workflow was implemented in a Google Colab-compatible Jupyter notebook to ensure it can be reproduced end-to-end without manual intervention.

Reproducibility Controls Used:

- Fixed random state (`random_state=42`) applied consistently across the train/test split and model training steps.
- Explicit preprocessing and modelling pipelines (`ColumnTransformer + Pipeline`) to guarantee identical transformations on train and test data.
- Hyperparameter search space (`GridSearchCV`) fully declared in code rather than manually tuned.
- Dataset loaded programmatically via KaggleHub, removing any dependency on manually placed local files.
- Markdown explanations placed before and after each code block to document intent, methodology, and interpretation.

10. Conclusion

This project delivered a complete, end-to-end churn prediction workflow — from raw data inspection through to a final, evaluated, and interpretable model. The analysis confirmed that customer engagement (active membership), age, geography, and account balance behaviour are the strongest churn signals in this dataset.

Key Results:

- A recall-focused Logistic Regression setup captures the large majority of churners (~77% recall), aligning with the project's success criterion of $\text{Recall} \geq 75\%$.
- A Random Forest model provides stronger overall ranking quality ($\text{ROC-AUC} \sim 0.86$), exceeding the $\text{ROC-AUC} \geq 0.80$ target and offering a good secondary option when precision matters more.
- Germany stands out as a high-risk market and warrants further business investigation into pricing, service quality, or competitive pressure.

Business Relevance:

The retention team can use the model's churn probability scores to prioritize outreach toward the highest-risk customers, rather than applying blanket retention campaigns. This targeted approach can reduce avoidable churn while controlling the cost of retention offers.

Limitations & Future Work:

- The dataset is static and does not include transaction history, complaint logs, or campaign response data.

- Future iterations could incorporate time-based behavioural features, tune the classification threshold against a specific retention budget, and benchmark additional algorithms such as Gradient Boosting.

11. References

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6. BITS Pilani. Advanced Apex Project-1 (M.Sc. DS & AI): Week-9 Final Submission Template.